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Can We Predict Male Strobili Production in *Araucaria angustifolia* Trees with Dendrometric and Morphometric Attributes?

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Abstract: Knowledge of the formation and correlation of reproductive structures with dendro/morphometric variables of the *Araucaria angustifolia* tree species is a tool for its conservation and viability for sustainable forest management. We counted visually in araucaria trees the number of male strobili in RGB images acquired by Remotely Piloted Aircraft System (RPAs) over forest remnants. The diameter at the breast height (d), total height (h), crown radii (cr), crown base height (cbh), periodic annual increment in d based on increment rolls were measured, and the morphometric indices and crown efficiency were calculated with these variables. The relationships of these variables with male strobili production were analyzed by Pearson's correlation and multivariate analysis techniques (cluster, factorial analysis, and main components). The morphometric variables correlated with the production of male strobili were d ($r = 0.58$, p-0.0002), crown diameter ($r = 0.62$, $p < 0.0001$), crown area ($r = 0.62$, $p < 0.0001$), coverage index ($r = 0.51$, p-0.001) and slenderness ($r = -0.39$, p-0.01). We argue that the production of male strobili is related to the vitality, dimension, density, growth space, and position in the stratum of the tree inside the forest, inferring a relationship between reproductive structures with the shape, size, growth space, and tree density. Such aspects shall be considered in future forest management initiatives in Southern Brazil.

Keywords: Brazilian pine; forest ecosystem; strobilus; mixed ombrophilous forest; silvicultural interventions

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1. Introduction

Araucaria or Brazilian pine (*Araucaria angustifolia* (Bertol.) Kuntze) is a dioecious species characteristic of the Mixed Ombrophilous Forest (MOF) in Southern Brazil and has initially occupied representative areas in the past [1]. As with all dioecious species, their regeneration depends on forming reproductive organs. The amount of these structures depends on the tree's age, crown size, density, light, climatic variables, the position occupied by the tree in the stratum, growth rate, and management of forest resources [2,3].

According to Wrege et al. [4] *A. angustifolia* occurs most frequently in the cold and humid climate mountainous regions of southern Brazil and, to a lesser extent, in the southeastern region, at average altitudes between 500 and 1000 m, although it can reach over 2400 m. In the region of the species occurrence, the average minimum temperatures in winter are 9.5 °C, and the average maximum temperatures in summer are 25.9 °C [5,6]. For

A. angustifolia, high average annual temperatures hinder the development and reproductive physiology of the species, harming its natural regeneration, dynamics, and conservation of the forest structure [7].

Thus, the reproductive organs exert a dominant influence on the structure of the forest, mainly because the *A. angustifolia* is the dominant species and is present in all strata of the forest. Management techniques for forming gaps, or originated by natural causes [8,9], help forest structure; spatial and temporal variations in light levels, canopy dynamics, soil nutrient and water regimes and maintain the characteristic uneven-aged nature of late-successional forest [10]. However, there are still few studies about the spatial distribution of male and female trees necessary for conserving this specific tree species, avoiding the risk of simplifying the structure, age distribution, and species composition [2,3,11].

It is expected that a high density of trees may harm the fertilization of female trees. The inflorescences are found at the end of the branches in adult female plants, and the gynostrobilus is composed of carpel leaves (megasporophyll) around a conical axis known as a cone. Androstrobili, also known, as male strobili are smaller in size, locally known as "mingotes", and they have an elongated axis and numerous scales inside pollen sacs, where pollen grains develop [12,13]. The reproductive process involves the formation of male strobili, the pollination by wind and the fecundation of the gynostrobilus to form the pine cone that contains edible seeds. Such seeds have strong ecological importance (food, fauna, species regeneration, gene flow). Interestingly, they also have importance in socioeconomic (family income, cuisine) and local culture (festivals and cultural events).

However, in isolated male trees, with a low plant density per unit area, the production of male strobili per plant is higher than in trees found inside forest remnants [14]. This aspect was most possibly due to less competition for light and soil nutrients. Furthermore, the distance between the trees also favors the permanence of the basal whorls for a longer time in the plant, as there is no friction and mechanical shocks with branches of neighboring trees as occurs in the closed forest. In this sense, the production of pine cones, in which the edible seeds are found, may increase with simple management practices to promote the development of the crown, as the control of the environment is a key factor for the tree to grow and produce in the desired way [15].

This study analyzes how shape and dimension variables correlate with the production of male strobili, as it consists of information for conserving the species and forest structure. Therefore, knowledge of morphometric variables is extremely important for forest management initiatives. Furthermore, the morphometric relationships and their interdimensional relationships are used to infer stability, vitality, and productivity, being a reference value for the planning of silvicultural interventions [16–21].

Due to the socio-economic and cultural importance of the targeted tree species and conservation of the mixed ombrophilous forest structure in Southern Brazil, the main objective of the study was to quantify and correlate the production of male strobili in isolated male trees with variables of crown shape and size with a view to the conservation and sustainable forest management of this tree species. This knowledge is important for better understanding the araucaria forest and insights into its structure, productivity, stability, resilience, and functioning of ecosystems [22–27].

Therefore, the study concludes with the answer to the tested hypotheses: (i) there is a correlation and significance of morphometry and crown efficiency in the production of reproductive structures in male *A. angustifolia* trees; (ii) variables of crown shape, density, and dimension are important in the production of male strobili; and (iii) male trees are important for the production of edible seeds in female trees and such production provides additional income of rural landowners.

2. Material and Methods

2.1. Study Site

The study area is a forest remnant with 84 ha of uneven-aged forest with *A. angustifolia* in the southern Santa Catarina state (Figure 1A). The main characteristic of the MOF, or

Forest with Araucaria, is the dominance in all strata of the *A. angustifolia* tree species, which forms a regular forest (Figure 1B). However, the current scenario does not present a mix with other tree species characteristic of this forest (Figure 1B) due to the absence of proper management initiatives.

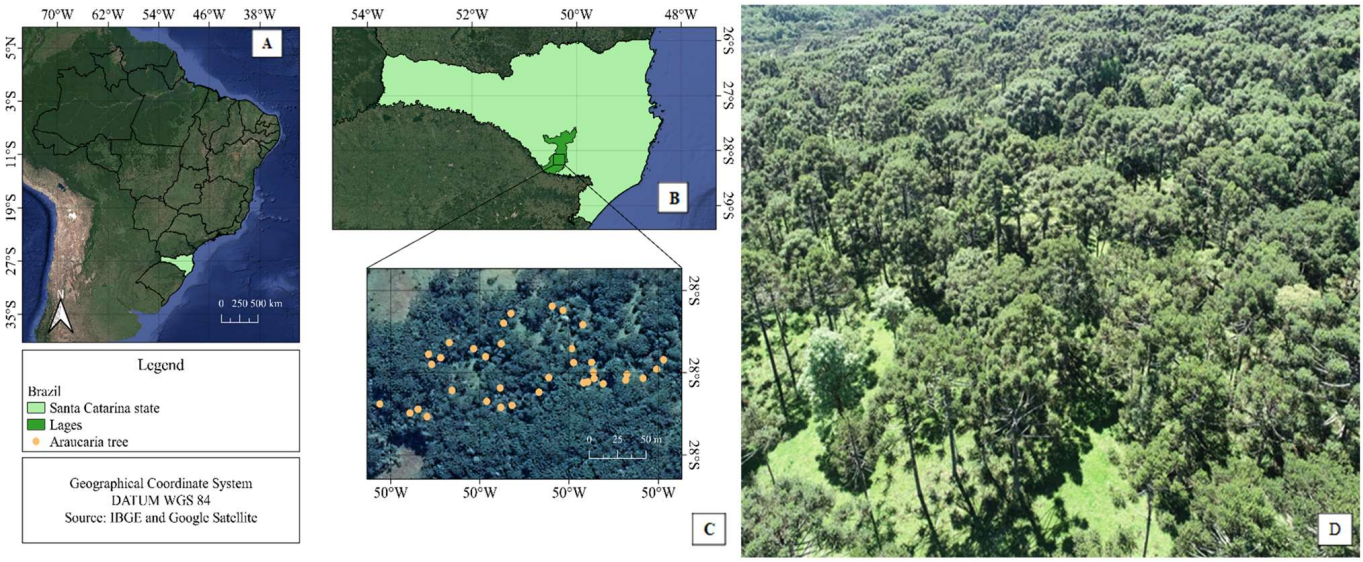


Figure 1. The geography of the study area: (**A**) study area location, (**B**) study area boundary, (**C**) shows each tree, location of the study area, and (**D**) aerial view of the study remnant, demonstrating the dominance of the *A. angustifolia* species and the formation of a regular forest.

The site's climate is Cfb temperate, according to the Köppen classification, and is constantly wet without a clear dry season. The altitude is 987 m a.s.l, with an average annual temperature of 15.2 °C and annual accumulated precipitation of 1684 mm [28].

2.2. Data Measurement

Thirty-six male *A. angustifolia* trees were selected based on the presence of reproductive structures (male strobili; the focus of the study) over different diametric classes and allowing to captured images of the crown using manual drone flight. The male trees were then tagged and georeferenced using handheld GPS (Garmin GPSMAP64x). Each male tree was considered a sampling unit. This number of trees meant 16% of sample trees for the hectare (ha). As the area has 330 trees.ha^{-1}, the proportion of male and female *A. angustifolia* trees is 70% and 30%, which corroborates previous forest inventories conducted in southern Brazil [2,3,29]. Female *A. angustifolia* trees usually show pine cones in their branches with varying sizes that are usually formed yearly. However, we do not consider the counting of female reproductive structures that are beyond the scope of this research.

For each male tree, the diameter at breast height at 1.30 m (d), total height (h), crown base height (cbh), and four crown radii (cr) were measured in the north, south, east and west directions, with the aid of a compass and Trupulse hypsometer.

Two increment rolls were removed at the d height (i.e., 1.30 m) to determine the annual periodic increment in d, considering a ten-year timeframe. Finally, the morphometric indices such as the crown length, crow surface area, slenderness, crown diameter, coverage index, crown efficiency and annual periodic increment in diameter (API_{10}) for ten years were calculated according to the following Equations:

$$cl = h - cbh \quad (1)$$

$$csa = \pi.\overline{rc}^2 \quad (2)$$

$$hd = {}^{h}/_{d} \quad (3)$$

$$API_{10} = (d - d_t)/t \quad (4)$$

$$cd = 2.\overline{rc} \quad (5)$$

$$ci = dc/h \quad (6)$$

$$CE_1 = API_d/csa \quad (7)$$

$$CE_2 = an/csa \quad (8)$$

where, cl: crown length (m); h: total tree height (m); cbh: crown base height (m); csa: crown surface area (m^2); $\overline{rc}$: average crown radius (m); hd: degree of slenderness; d: diameter at breast height (cm) in 2019; API_{10}: annual periodic increment in diameter ($cm.year^{-1}$) for a period of ten years; d_t: diameter at breast height (cm) obtained from the average of the two increment rules at the beginning of the period (i.e., 2009); t: period of time considered; cd: crown diameter (m); ci: coverage index; CE: crown efficiency; an: number of male strobili. CE_1 and CE_2 are two different proposed ways to evaluate crown efficiency. While CE_1 consider an average for a timeframe of ten years, csa is a valid measurement for a given year (i.e., 2019), and CE_2 is the average of two consecutive years (i.e., 2019 and 2020).

The counting of male strobilus for each tree was carried out by visually inspecting and labeling RGB images acquired by the DJI Phantom 4 Pro V2.0 with an attached camera of 20 Mpixels. Then, a manual flight was performed to explore the tree's total crown considering with a mechanical shutter and 8.8 mm focal length. In order not to count the same strobilus more than once, the images were taken on each axis following the cardinal directions (north, south, east and west), as well as from the top of the tree, obtaining at least five images of every single tree. Male strobili were quantified for the same sampled trees for two consecutive years, i.e., during November and December 2019 and 2020, respectively. These were the preferable months that male strobili were in the maturation stage in the study area. Although its development occurs from February to August, its maturation occurs between September and November and may vary in time in other environments in its occurrence area in Southern Brazil. Usually, when immature, its color is green, and when reaching maturity, yellow to brown [30,31], making it easier to distinguish from the needles (i.e., leaves), which is why they were chosen for the quantification in this research (Figure 2C,D).

Depending on the objects' size, the aerial cover's height (distance) was not greater than 30 m to ensure sub-centimeter quality (i.e., 0.8 cm). The identification and quantification of male strobili occurred through a visual interpretation of the images, using the VGG Image Annotator Platform as a facilitator. The quantification was then performed manually and double-checked by two users to minimize the eventual double counting of the structures in two sequential images overlapping (Figure 2A,B). Since the counting was done manually by visual inspection of the images, the obtained results were considered a reference. It is important to highlight that the presence of eventual dry needles (i.e., leaves) may confuse the interpreter in the labeling process. It requires then several zooms in the image to detect the object structure. Therefore, for this research, we did not explore other digital image processing techniques to count these objects automatically since they were also beyond the scope of this study.

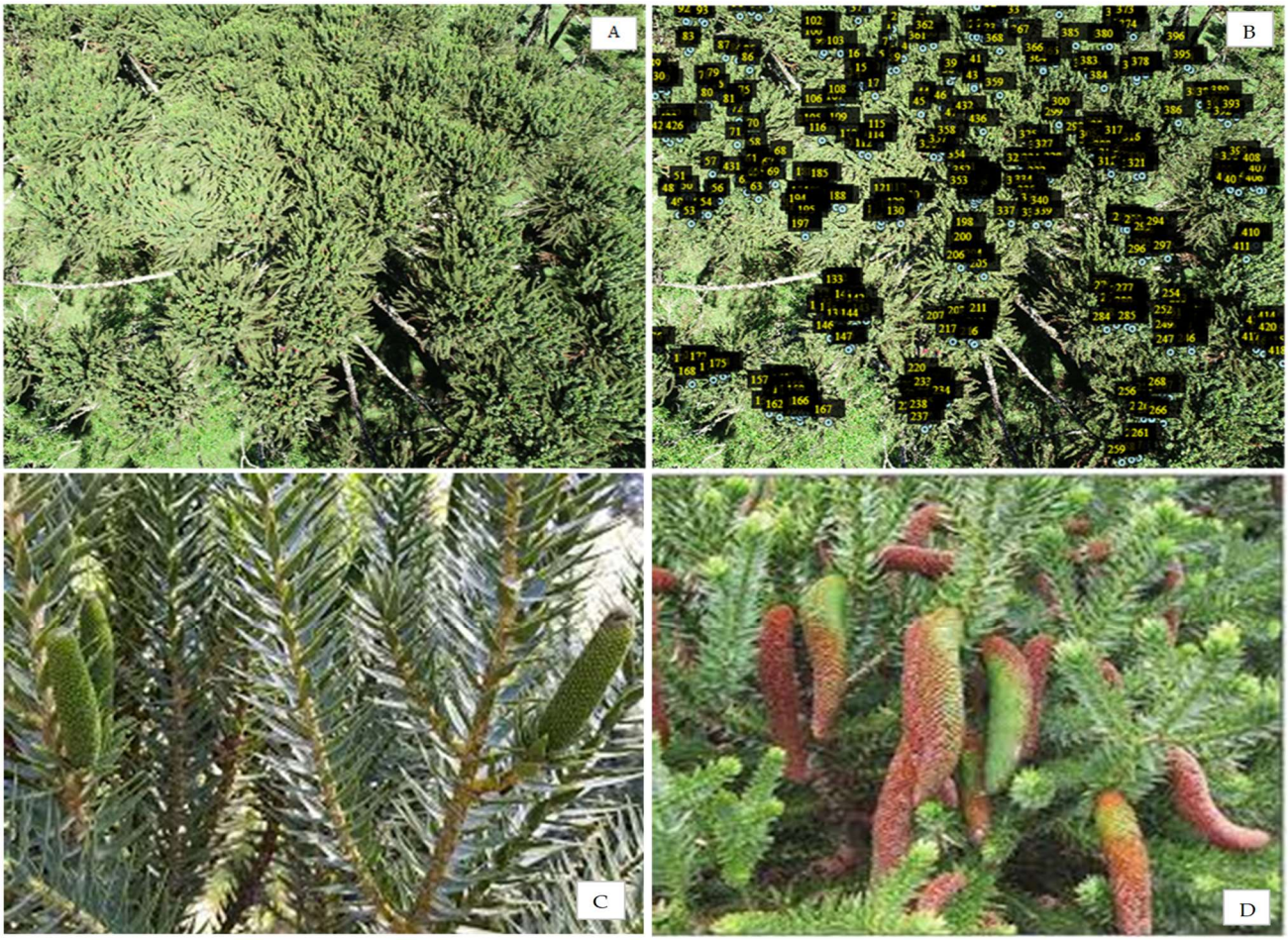


Figure 2. (**A**) The subset of an aerial image of the *A. angustifolia* crown obtained with controlled drone flight; (**B**) the spatial distribution of male strobili on the same subset using VGG Image Annotator Platform; (**C**) immature male strobili and (**D**) mature male strobili. Source: Authors (**A**,**B**); Katia Pichelli1 (https://www.minutorural.com.br/noticia/1217/pesquisa-desenvolve-araucarias-macho-que-produzem-polen-em-menos-da-metade-do-tempo. Accessed on 10 October 2022.) (**C**), Wendling and Zanette [30] (**D**). A perspective view of the study area and the crown structure is shown in Figure 1.

Afterward, we submitted the data to the Pearson's correlation analysis to predict the production of male strobili (an) as a function of morphometric and dendrometric variables. After, the *p*-value was assessed to check whether a correlation is statistically significant. The adjustments were made to generalized linear models regression (Equations (9)–(11)).

$$an = b_0 + b_1 * d + \varepsilon_i \quad (9)$$

$$an = b_0 + b_1 * csa + \varepsilon_i \quad (10)$$

$$an = b_0 + b_1 * hd + \varepsilon_i \quad (11)$$

where, an: number of male strobili; d: diameter at breast height (cm); csa: crown surface area (m^2); hd: degree of slenderness; and ε_i: random error.

The equations' accuracy were analyzed by the statistics of deviance Akaike's and Bayesian information criterion and random errors. Graphical analyses of the observed and estimated data were then performed. Insights into the results and their importance in terms of species conservation forest structure were also highlighted. Complementary, the data were also submitted to multivariate analysis techniques such as cluster, factorial,

and principal component analysis for a better understanding of the relationship of both dendrometric and morphometric attributes with male strobili production.

To avoid the effect of arbitrariness, the data were standardized so that one variable does not interfere with another and contribute equally to the analysis. All analyzes were performed using Statistica 7 (Palo Alto, CA, USA) software [32].

The above procedures allowed us to establish the relationship of male strobili with dendrometric and morphometric variables. Since the targeted tree is a dioecious species, seed formation depends on pollination and fertilization of reproductive structures in female trees, forming pine cones that contain edible seeds. Besides local economy, such edible seeds are important for species regeneration, conservation, forest structure, and fauna maintenance.

Therefore, we analyze the importance of the results with the gross production of edible seeds marketed in the three states of southern Brazil where this tree species occurs. The official information on the products was obtained from the official repository of the National Company for Food Supply [33]. We focus more specifically on the 18 municipalities of the Association of Municipalities of the Mountain Region, Santa Catarina (AMURES) where the study area is located.

3. Results

The descriptive statistic of the dendrometric, morphometric, number of male strobili, annual periodic increment in diameter, and crown efficiency are presented in Table 1. Here, we did not observe ingrowth and mortality. Figure 3A shows the number of male strobili per year, and d class center.

Table 1. Descriptive statistics of dendrometric, morphometric, and number of male strobili for *Araucaria angustifolia* in the study site for two consecutive years, south Brazil.

Variable	Mean	Minimum	Maximum	Standard Deviation
an	317	80	885	180.2
d	40.1	16.4	69.7	9.0
h	14.2	8.5	18.7	2.2
API_{10}	0.56	0.24	0.98	0.20
cd	9.57	4.7	19.6	3.28
csa	80.21	17.35	301.72	59.71
ci	0.68	0.36	1.17	0.21
hd	39	21	84	14
cl	6.9	2.7	12.2	2.23
CE_1	0.01042	0.00110	0.03558	0.00771
CE_2	5.03287	1.92449	11.39336	2.13079

an = number of male strobili; d = diameter at breast height (cm); h = height in (m); API_{10} = annual periodic increment in diameter ($cm.year^{-1}$); cd = crown diameter (m); csa = crown surface area (m^2); ci = coverage index; hd = degree of slenderness; cl = crown length (m); CE_1 = crown efficiency ($cm/year/m^2$); CE_2 = crown efficiency (an/m^2).

Pearson's correlation coefficients showed significant values for the dendrometric and morphometric variables and their p-value, in order: d ($r = 0.58$, p-0.0002); cd ($r = 0.61$, $p < 0.0001$); csa ($r = 0.62$, $p < 0.0001$); ci ($r = 0.51$, p-0.0013); ge ($r = -0.39$, p-0.0196). Thus, the production of male strobili was shown to be directly proportional to the variables d, cd, csa and ci and inversely proportional to the relationship hd. Furthermore, the number of productive structures increases with the size and shape of the trees (Figure 3B,D) and decreases with increasing density and slenderness value (Figure 3C).

The number of male strobili (Equations (9)–(11)) obtained a lower value of deviance, Akaike, and Bayesian information criterion, standard error and probability of coefficients <0.0001 (Table 2). Therefore, the models show accuracy without discrepant values. Interestingly, the production of male strobiles is associated with an increase in crown size (cd) and canopy surface (csa). However, not linearly with increasing diameter, but concentrated in

trees from 30 to 60 cm, linked to age, physiological capacity, position the tree occupies in the stratum, and density, as old trees, in competition, reduce efficiency and production.

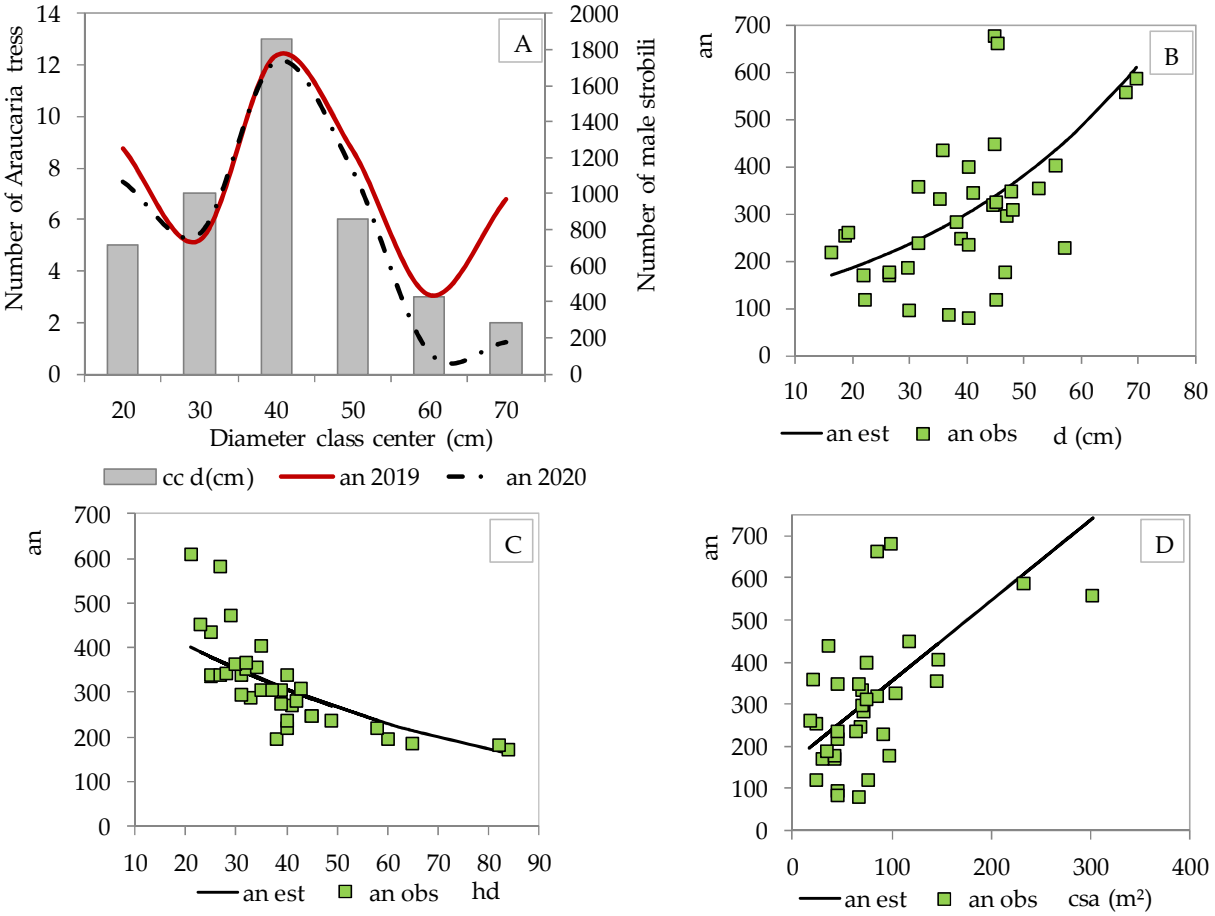


Figure 3. (**A**) Number of male strobili (an) counted per diameter class and total for the two sampling periods (2019 and 2020, respectively); (**B**–**D**) accuracy of the adjusted models for *Araucaria angustifolia* trees for the relationship between male stobili production and dendrometric and morphometric variables.

Table 2. The equation for the number of male strobili in *A. angustifolia* trees as a function of dendrometric and morphometric variables and their statistical fitted.

Equation	b_0	SE	b_1	D	AIC	BIC	LF
9	4.7573 *	0.24	0.0237 *	7.5	456.7	461.4	G–$ln(\mu)$
10	163.8949 *	36.12	1.899 *	7.4	456.2	460.9	$G(\mu)$
11	6.2944 *	0.21	−0.0143 *	9.4	464.7	469.4	G–$ln(\mu)$

b_0 and b_1 = coefficients; * = Chi-square probability test, all coefficients—Pr > ChiSq < 0.0001; SE: standard error; D: deviance; AIC = Akaike's information criterion; BIC = Bayesian information criterion; LF ($G(\mu)$) = identity link function, gamma distribution; G–$ln(\mu)$ = logarithmic link function, gamma distribution.

This was evident in the relationship between male strobile production and the degree of slenderness (Figure 3C). In addition, we also noticed a higher value for the degree of slenderness, higher density, competition for space and light, smaller crown dimension, trees in the codominant, dominant stratum, and lower amount of male strobili. Thus, the above findings seem to be essential information when considering the viability of management and conservation plans for the species.

It is noted that Figure 3A shows seasonality in annual production and the relationship of production to crown size and tree d. The low number of trees in smaller d classes is because the age of production of male reproductive structures occurs only between 12

and 15 years old [34], while in trees of larger d and age the production decreases due to lower physiological capacity. For 2019, the number of male strobili counted was 6406 for the total of sampled trees. For 2020, 5000 male strobili were counted, totaling 11,406 male strobili in these two consecutive years (Figure 3A). Lower production of male strobili may indicate lower pollination and, consequently, a lower amount of pine cones, edible seeds, and natural regeneration of the species.

The trees with the highest number of male strobili for the two consecutive years were those of d classes between 35 and 45 cm. Interestingly, these are consequently the largest crown diameter and crown surface area (Figure 3). Furthermore, the results show that the greater the number of male strobili, the greater the crown efficiency, crown diameter, increment rate, and tree growth conditions (Table 1, Figures 3 and 4).

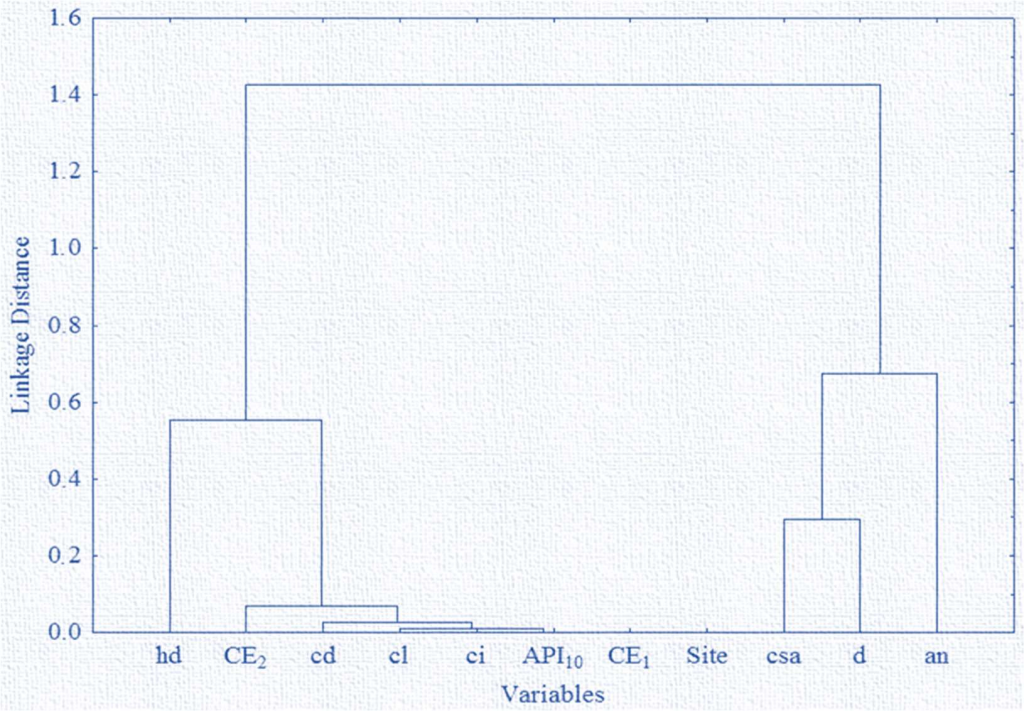


Figure 4. Dendrogram for dendrometric and morphometric grouping and related to male strobili production, which: cl = crown length; hd = degree of slenderness; CE = crown efficiency; API_{10} = annual periodic increment in diameter; ci = coverage index; csa = crown surface area; cd = crown diameter; d = diameter at breast height; an = to male strobili number.

Figure 4 shows the cluster analysis and the formed dendrogram with their respective groups and relationships. The vertical scale indicates the level of similarity between male strobili and their relationship with the trees' shape and size variables. The horizontal axis shows the grouping in order of influence on the production of male strobili. The influence of the variables informed by the two groups shows the importance of the growth space, density, positioning in the stratum, and crown dimension as important factors in the formation of male strobili [2,3,35,36].

Factor analysis (Table 3) showed four components, two of which explained 90.65% of the variation in male strobili formation and were used to extract the principal components. In addition, the correlation analysis in the factor analysis showed higher values (r = 0.9), indicating a strong correlation of (male strobili) with dendrometric and morphometric variables.

Table 3. Eigenvalues and percentages of variance are explained for each component.

Component	Eigenvalue	% Total Variance	Cumulative Eigenvalue	Cumulative %
1	6.67	66.67	6.67	66.67
2	2.4	23.99	9.07	90.66
3	0.55	5.54	9.62	96.20
4	0.31	3.01	9.93	99.25

Factor 1 explained the relationship between the number of male strobili and the variables API_{10}, cd, ci, cl, $CE_{1,}$ and CE_2. Factor 2 with the variables d and csa. The results indicate that the formation of male reproductive structures is directly linked to crown development, growth, and efficiency.

After the formation of reproductive structures, their permanence in the tree until the dissemination of pollen will strongly depend on weather conditions such as wind, precipitation, relative humidity, radiation, and the occurrence of climatic phenomena (storms).

The distribution of resources between trees is difficult to measure, and the growth rates are proportional or absolute. However, the growth-dimension relationships can be used as proximity to the resource-dimension relationship [37], and this study expresses relationships physiologically as the formation of reproductive structures.

Figure 5 shows the ordering diagram of the original variables of the four main components. Again, the morphometric, growth, and efficiency variables are superimposed (left side of the circle), demonstrating a similar representation in the number of male strobili. Interestingly, the variables d, csa and hd are close to the unit circle, denoting a greater relationship with the number of male strobili.

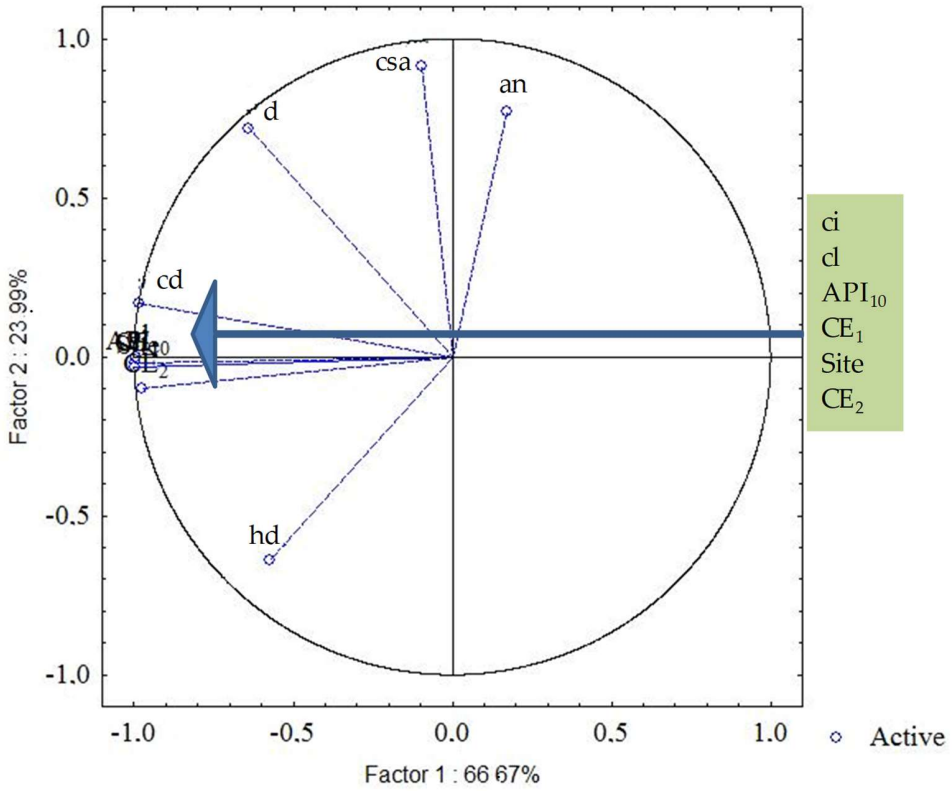


Figure 5. Circle of correlations between number of male strobili and dendrometric and morphometric variables for *A. angustifolia*. The blue arrow and the green frame on the right indicate the original variables grouped in the principal components circle, which: cl = crown length; hd = degree of slenderness; CE = crown efficiency; API_{10} = annual periodic increment in diameter; ci= coverage index; csa = crown surface area; cd = crown diameter; d = diameter at breast height; an = male strobili number.

The results with the application of multivariate analysis techniques show that trees' shape and dimension components are preponderant for the formation of male strobili. They also indicate the need to manage competition, density, space and growth to reconcile wood production and other ecosystem structures and services [38], such as reproductive organs, regeneration, and biodiversity.

4. Discussion

The tree crown size seems to be a key variable in this research as it correlates with the space a tree occupies and the physiological tree functions (Figures 4 and 5). For instance, crown projection area and volume can be proxy variables for leaf area index and biomass [39,40].

In addition, tree crown and tree stem diameter growth are very competitively sensitive [41]. These authors also mentioned that several tree characteristics, functions, and services can be derived from vertical and lateral crown expansion (i.e., height, height to crown base, and diameter).

This research combined dendrometric and morphometric variables to describe their influences and correlations on male strobili formation. Strong relationships between the variables showed that male reproductive structures depend on the crown and size of the tree. Similar results were reported by Hess et al. [2] and Atanazio et al. [3] when analyzing the formation of female structures in *A. angustifolia* trees.

In summary, the formation of reproductive structures and the conservation of the tree species depend on crown development and growth. Therefore, it can be said that fewer reproductive structures lead most probably to fewer edible seeds, which also affects tree species' regeneration. For example, *A. angustifolia* is usually propagated by its seeds, which have low percentages of germination once they have been stored and are classified as recalcitrant [42].

Costa et al. [43] demonstrate that genetic composition and mating system estimates of different cohorts may change yearly. The temporal variation demonstrates the importance of driving multiyear analyses, especially for long-lived tree species, and recommends that covering the existing temporal and spatial variation is especially relevant for future research and seed collection plans involving *A. angustifolia*. This is because the generated information has direct implications for management and conservation strategies for this species and, consequently, for the ecosystem in which it occurs.

The statistical response confirms the hypotheses tested, bringing good insights for tree specie conservation, regeneration promotion, and moderate silvicultural interventions. Besides assuring a better crown development due to a higher space available, such initiatives assure ecosystem benefits and services, as well as economic profitability and social development for producer families who depend on selling edible seeds from pinecones. Furthermore, forests offer a solution to climate change through carbon storage and providing ecosystem services and sustainable products. While reducing greenhouse gas emissions are required to address the climate change crisis, restoring and expanding forests increases terrestrial carbon sinks. Therefore, it is critical to manage our forests sustainably. This requires innovative and interdisciplinary research and practices [44].

According to CONAB [33] the national production of edible seeds in 2016 totaled 7.7 thousand tons, emphasizing that a decrease of 7.7% was mentioned from the previous year. The decrease is explained by alternating production cycles with years of counter-harvest after two or three consecutive years of high edible seed production. However, production remains lower, with a 20% reduction in some municipalities in the southern states of Brazil in 2022. The explanation is that after a vast production, this species produces little again as it becomes exhausted, alternating good productivity with less intense periods.

However, this explanation reported by CONAB [33] is not scientific and is a subject that requires more research. Seasonality is due mainly to climate changes, and exhaustion is a physiological capacity. Nevertheless, the results show that the formation of reproductive

structures is strongly related to crown size, d, space, and efficiency, which can be obtained with forest management practices.

On the other hand, no silvicultural interventions have been allowed for the last three decades due to restrictive legislation, contributing to the local disinterest in increasing new forest areas with *A. angustifolia*. Currently, the only income resource is the sale of edible seeds, which also puts inevitable pressure on the remnants since fewer seeds are left for regeneration due to overexploitation, which may start a vicious cycle. According to Wendling [45] the unsustainable exploitation of this species, caused by the ever-increasing demand for *A. angustifolia* edible seeds (nuts), seriously threatens the regeneration of this specific tree species.

In the 18 municipalities of AMURES, several families have profited from harvesting pine cones and selling their edible seeds. The report published by Dornelles [46] also mentioned that from the 18 municipalities, the sale value in 2019 was BRL 3,939,895.74 (million). These values are those declared in invoices. However, they may be different from reality because edible seeds are sold directly to consumers in an informal market. Considering that around three thousand families live in rural properties, this represents a profit of around BRL 1300.00. The official quotation of the USD by the Brazilian Central Bank fluctuated from BRL 3.8595 and BRL 4.0307 from January to December 2019. However, this income could be supplemented with the sale of timber logs under sustainable management practices. Based on previous studies, the monthly income from wood exceeds BRL 15,000 (thousand), when considering a timeframe of 20 years [47].

The results indicate crown shape and size strongly correlate with male *strobili* production. Thus, silvicultural actions are necessary to promote light capture, as this resource is a vital factor affecting tree growth, survival, and reproduction [48]. Furthermore, the characteristics of the crown and leaves also interfere with the amount of light captured and the efficiency of the use of this resource and contribute to understanding the strategies of the growth, structure, and dynamics of forest communities [49].

Thus, in addition to providing conditions for the availability and use of resources, sustainable management of *A. angustifolia* is strongly suggested [42,50,51]. However, the absence of silvicultural interventions in the forest will most probably contribute to the reproductive decline, mainly because the remnants of this forest have dominant trees, which are large and reach high longevity, stagnant growth, and low natural regeneration [16,18,29,52]. The result is a non-dynamic forest structure due to the legal restrictions that have remained unchanged for decades. Consequently, a lower diversity in the number of tree species and vertical strata is observed [16,18,49,53].

The results indicate that the crown dimension, crown area increment, and efficiency are strongly related to vigor, tree position in the stratum and light capture in forming of reproductive structures [2]. Plant growth rates are driven by factors that influence resource availability (light, nutrients and water), the number of resources captured and the efficiency of resource use. Focusing on light resources, the amount of light captured and resource use efficiency in trees are mainly determined by crown characteristics and leaf traits [54].

In this sense, crown attributes (shape, dimension, and architecture) and spatial organization (density, structure) determine the growth rate and formation of reproductive structures (dendrogram of variables in Figure 4). The most significant variables related with the number of male strobili were crown size, diameter and density, represented by the hd relationship. The study motivates further forest management efforts for multiple forest ecosystem functions and services.

As previously commented by Costa et al. [51], advancements in managing mixed forests with *A. angustifolia* trees bring new perspectives for both management and conservation initiatives. However, many other factors still need to be considered in the context of mixed forest management, especially those related to the ecology of systems that are beyond the scope of this research. Therefore, further studies are suggested to ensure the proper management protocols of this particular tree species.

This research also shows the applicability of high spatial resolution images acquired from RPAs to explore the production of male strobili. Further steps could evaluate the applicability of prediction methods to count male reproductive structures allowing complementary studies relating them to climate variables and dendrochronology, such as those experiments conducted recently by Hadad et al. [55] in a similar tree species in south Argentina and Chile (*A. araucana*). It also brings new perspectives for combining photogrammetric techniques and checking the applicability of digital image processing and deep learning (DL) techniques to count reproductive structures more efficiently and also to extract relevant biophysical attributes such as the crown from the images. It also eliminates any possibility of counting twice a single structure in sequential images with overlapping. Hence, it allows further analysis of the robustness of the reported approach to count objects and compare it with forthcoming up-to-date methods.

Interestingly, it would be to evaluate hyperspectral sensors or cameras acquiring images in additional wavelengths, such as red-edge and near-infrared, to evaluate the counting of these structures in the previous maturity phases. However, some challenges may mainly occur since the typical green color of these reproductive structures is very similar to the green color of their needles (e.g., leaves), bringing some challenges and encouraging further studies. Therefore, both the marking point, annotated in the center of each structure and a bounding box annotation using bounding boxes as suggested by Biffi et al. [56] in commercial apple fruit orchards is recommended here for completeness of comparisons allowing time-demanding performance assessments.

Interestingly, it would be to study the spatial distribution of male and female trees in specific forest remnants and to count pine cones in female trees that contain edible seeds for harvesting optimization and ecological studies. The *A. angustifolia* is an important tree species that can be used for recovering degraded environments in southern Brazil [57].

5. Conclusions

The results of this research confirmed the proposed hypotheses. First, there is a significant correlation between the number of male strobiles per tree with dendrometric and morphometric variables (i.e., density, crown efficiency, age, and physiological capacity of the tree). Second, the production of male strobili is directly proportional to the diameter at the breast height (d), shape, and size of the crown. Interestingly, the production of male strobili is also proportional with the upper dimensions corresponding to advanced age (i.e., old trees) and young trees, smaller in diameter and not physiologically mature, and were inversely proportional to density, position in the stratum, and competition. Third, it was also possible to relate the importance of the formation and production of male strobili with edible seeds, functions, and ecosystem services, both its structure and, in the case of araucaria, the local economy, which has decreased its production and income continuously since 2017. Therefore, more studies and public initiatives are encouraged to manage and conserve this important tree species.

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